

# Standard Flips of Discrepancy One: Extremal $J$ -Normalization and the Meijer Aperture at $\nu = 1$

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## Abstract

Shen and Shoemaker compute the extremal quantum spectrum of a standard flip  $X \dashrightarrow X'$  with exceptional locus  $\mathbf{P}(V) \subset X$ ,  $\text{rank } V = r$ ,  $\text{rank } V' = s$ , and show that the Gamma-class decomposition of  $H^*(X)$  attached to the Belmans–Fu–Raedschelders semiorthogonal decomposition is a decomposition into asymptotic classes. Their Theorem 4.4, which identifies an explicit hypergeometric series with the extremal  $J$ -function of the local model, assumes  $r - s > 1$ ; their Remark 4.5(3) asserts that for  $r - s \leq 1$  the series is not  $J$ -normalized; and the Barnes asymptotic expansion of their Section 7 is applied under the same inequality. The printed dependency chain of their Theorems 1.2, 1.4 and 9.14 therefore does not reach the discrepancy-one range  $r = s + 1$ ,  $s \geq 1$ , which contains every codimension-two blow-up.

We supply the two missing steps. First, the degree- $d$  summand of their formula (35) has  $z$ -order at most  $1 - s - (r - s)d$ ; for  $r = s + 1$  and  $s \geq 1$  this is at most  $-1$  for every  $d \geq 1$ , so the series is  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ , no mirror-map correction arises, and the cone membership recorded in Remark 4.5(3) identifies the series with the extremal  $J$ -function. The only remaining formal failure of that normalization is the degenerate endpoint  $s = 0$ , whose point fibres contain no extremal line. Second, at  $\nu := r - s = 1$  the sector printed after their Lemma 7.4 is unavailable, being the  $\epsilon = 1$  case of their own Theorem A.1; the correct  $\epsilon = 1/2$  sector still meets the sector of their Proposition 8.2 in an open sector of opening  $2\pi$  that contains both the nonzero-eigenvalue ray and the tame ray. With the reductions of their Sections 9.1–9.4 unchanged, this extends Theorems 1.2, 1.4, 9.14 and Corollary 1.5 to every standard flip with  $r = s + 1$  and  $s \geq 1$ .

**Keywords.** standard flip; quantum spectrum; Gamma class; asymptotic class; Meijer  $G$ -function; semiorthogonal decomposition.

## 1 Introduction

A *standard flip*  $X \dashrightarrow X'$ , in the setting of Bondal–Orlov and of Belmans, Fu and Raedschelders [3], is built from the following data. There is a birational morphism  $p: X \rightarrow \bar{X}$  of smooth projective varieties, an isomorphism away from  $F = p^{-1}(Z)$  for a smooth  $Z \subset \bar{X}$ , and vector bundles  $V, V' \rightarrow Z$  of ranks  $r$  and  $s$  with

$$F \cong \mathbf{P}(V), \quad N_{F|X} \cong \pi^*(V') \otimes \mathcal{O}_{\mathbf{P}(V)}(-1),$$

where  $\pi: F \rightarrow Z$ . Blowing up  $F$  and contracting the exceptional divisor  $E \cong \mathbf{P}(V) \times_Z \mathbf{P}(V')$  in the other direction produces  $X'$ , which contains  $F' \cong \mathbf{P}(V')$  of codimension  $r$ .

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Throughout,  $r > s$ . Since  $F \subset X$  has codimension  $s$  and  $F' \subset X'$  has codimension  $r$ , the two blow-up formulas for the canonical class on  $\hat{X} = \text{Bl}_F X$  give  $\tau^* K_X - \tau'^* K_{X'} = (r - s)E$ . We call

$$\nu := r - s$$

the *discrepancy* of the flip;  $s = 1$  is the blow-up of  $\bar{X}$  along  $Z$ , and  $s = 0$  is the degenerate case  $X = F = \mathbf{P}(V)$ ,  $X' = \emptyset$ , of a projective bundle.

Shen and Shoemaker [13] compute the spectrum of quantum multiplication by  $c_1(X)$  after restricting the Novikov parameter to the extremal ray contracted by  $p$ , and prove that the Gamma-class decomposition of  $H^*(X)$  induced by the semiorthogonal decomposition of  $D^b(X)$  of [3] is a decomposition into strong asymptotic classes in the sense of Galkin–Golyshev–Iritani [7]. This is [13, Theorems 1.2, 1.4 and 9.14, Corollary 1.5], and it is stated for standard flips with the explicit remark that it covers the blow-up case  $s = 1$  and the projective-bundle case  $s = 0$ .

The proof passes through an explicit hypergeometric series. In the local model  $T = \text{tot}(\pi^* V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$ , after the extremal specialization [13, (33)] of Novikov and parameter variables, [13, Theorem 4.4] states that the extremal  $J$ -function of  $T$  is

$$J^T(q, t, z) = z e^{t/z} q^{c_1(T)/z} \sum_{d \geq 0} q^{(r-s)d} \frac{\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)}{\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)}, \quad (1)$$

which is [13, (35)]. Everything after Section 4 of [13] runs on this formula: the ring presentation [13, (37)] is deduced from it, the extremal spectrum is read off from that presentation, and the modified  $J$ -function of [13, (39)–(40)] used in the asymptotic analysis of Sections 7–9 is defined from it.

Three hypotheses in the printed argument constrain  $\nu$ . [13, Theorem 4.4] is stated under  $r - s > 1$ . [13, Remark 4.5(3)] asserts that for  $r - s \leq 1$  the right-hand side of (1) is not of the form  $z\mathbf{1} + (\text{lower order in } z)$  and therefore cannot equal the  $J$ -function of  $T$ , although it does lie on Givental’s Lagrangian cone of  $T$  and so is an  $I$ -function. Finally, the Barnes asymptotic expansion of the Meijer  $G$ -function of [13, Lemma 7.4] is introduced by the sentence “For  $r - s > 1$ ”, and the sector [13, (64)] printed with it is derived under that inequality. Under the standing assumption  $r > s$ , the range excluded by these three hypotheses is exactly  $\nu = 1$ , that is

$$r = s + 1.$$

It contains every codimension-two blow-up, the case  $(r, s) = (2, 1)$ . So the dependency chain printed for [13, Theorems 1.2, 1.4 and 9.14] does not literally reach the discrepancy-one range, even though the statements are made for standard flips and blow-ups without a restriction on  $\nu$ .

This note supplies the two missing steps. Both are internal to [13]: the first uses only (1) and the cone membership already recorded in [13, Remark 4.5(3)], and the second uses only [13, Theorem A.1], in the form the authors state it.

**Theorem 1.1.** *Let  $r = s + 1$  with  $s \geq 1$ . Then, in the extremal specialization [13, (33)] and with the same formal variables and cohomological completion as in [13, Theorem 4.4], the degree- $d$  summand of (1) has  $z$ -order at most  $-1$  for every  $d \geq 1$ . Consequently (1) equals  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$  with  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$ , and equals the extremal  $J$ -function of  $T$ . In particular the presentation [13, (37)] of [13, Theorem 4.6] holds for  $r = s + 1$ ,  $s \geq 1$ , as does [13, Theorem 1.2], with the single nonzero eigenvalue  $\lambda_0(q) = (-1)^s q$  of multiplicity  $\dim H^*(Z)$ .*

The negative assertion of [13, Remark 4.5(3)] is therefore accurate, in the range  $r > s$  it addresses, only for  $s = 0$ , where  $(r, s) = (1, 0)$  and the degree-one summand of (1) does contribute in  $z$ -order zero.

**Theorem 1.2.** *Let  $r = s+1$  with  $s \geq 1$ , and let  $0 \leq k \leq 1$ . The asymptotic expansion of [13, Proposition 7.5] for the component indexed by  $m = -k$  is valid on  $|\arg(z/q) + (2m+s)\pi| < \frac{3\pi}{2}$ , and not on the wider sector obtained by setting  $r-s=1$  in [13, (64)]. The expansion of [13, Proposition 8.2] is valid on  $|\arg(z/q) - (1-s)\pi| < \frac{3\pi}{2}$ . These two sectors meet in an open sector of opening  $2\pi$  containing both the eigenvalue ray  $\arg(z/q) = -(2m+s)\pi$  and the tame ray  $\arg(z/q) = (1-s)\pi$ .*

For a codimension-two blow-up,  $(r, s) = (2, 1)$  and  $k = m = 0$ , the two sectors are  $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$  and  $|\arg(z/q)| < \frac{3\pi}{2}$ , and they meet in  $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ , which contains the tame ray 0 and the eigenvalue ray  $-\pi$ .

**Corollary 1.3.** *[13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for every standard flip with  $r = s+1$  and  $s \geq 1$ , in particular for every codimension-two blow-up.*

The mechanism behind Theorem 1.1 is a single count, and it is worth saying where the natural first count goes wrong. The numerator of the degree- $d$  summand of (1) has  $sd$  factors and the denominator has  $rd$  factors. Treating all of them as linear in  $z$  gives  $1 + sd - rd = 1 - \nu d$  for the maximal  $z$ -order, which is 0 at  $d = 1$  when  $\nu = 1$ : a curve-induced contribution in  $z$ -order zero, hence a nontrivial mirror map, hence exactly the failure asserted in [13, Remark 4.5(3)]. But the inner index in the numerator runs over  $0 \leq m \leq d-1$ , so  $s$  of its factors are  $\sigma_j - H$ , free of  $z$ . The correct maximal order is  $1 + s(d-1) - rd = 1 - s - \nu d$ , and at  $\nu = 1$  the  $s$  missing powers of  $z$  are precisely what pushes every  $d \geq 1$  summand below  $z^0$  as soon as  $s \geq 1$ . Once no  $z$ -order-zero contribution survives, the  $z^0$  coefficient of (1) is the parameter  $\bar{\mathbf{t}}$  itself, the mirror map is the identity, and uniqueness of the  $J$ -slice of the cone converts the cone membership of [13, Remark 4.5(3)] into an identification.

Theorem 1.2 is not a strengthening but a correction of an aperture. The Meijer  $G$ -function of [13, Lemma 7.4] has  $\nu = r - s$ , and [13, Theorem A.1] — Barnes' expansion in the form of Meijer [10] and Luke [9] — is valid on  $|\arg t| < (\nu + \epsilon)\pi$  with  $\epsilon = 1$  for  $\nu > 1$  but only  $\epsilon = \frac{1}{2}$  for  $\nu = 1$ . The sector [13, (64)] is the  $\epsilon = 1$  sector; substituting  $r - s = 1$  into it claims an aperture  $\pi/2$  wider on each side than Barnes' expansion supports. The point of Theorem 1.2 is that the smaller, correct sector is still large enough: the common sector with [13, Proposition 8.2] retains opening  $2\pi$ , so the oriented comparison of the nonzero-eigenvalue and tame expansions that the argument of [13, Section 9] performs is available at  $\nu = 1$  as it is for  $\nu > 1$ .

What is not covered is the formal endpoint  $(r, s) = (1, 0)$ , the rank-one projective bundle, where  $\mathbf{P}(V) = Z$ ,  $X' = \emptyset$ , and the point fibres contain no line class  $[L]$ . Its displayed hypergeometric expression also fails the required order count. Nothing here bears on  $\nu > 1$ , where the argument of [13] already applies, and nothing here changes any statement of [13] outside the two hypotheses named above.

Section 2 fixes notation and records the three places where  $\nu$  is constrained in [13], together with what depends on each. Section 3 proves Theorem 1.1, Section 4 proves Theorem 1.2, and Section 5 assembles Corollary 1.3 and states the boundary of the repair.

*Convention.* Numbered items of [13] are always cited explicitly, as in [13, Theorem 4.4] or [13, (35)], and refer to the version arXiv:2502.08762v2. Displays numbered here, such as (1), are our own.

## 2 The extremal setting and the three constraints on $\nu$

We keep the notation of [13]. Let  $H = c_1(\mathcal{O}_{\mathbf{P}(V)}(1))$ , let  $\rho_1, \dots, \rho_r$  and  $\sigma_1, \dots, \sigma_s$  be the Chern roots of  $V$  and  $V'$  pulled back to  $\mathbf{P}(V)$ , and let  $T = \text{tot}(\pi^*V' \otimes \mathcal{O}_{\mathbf{P}(V)}(-1))$  be the local model, so that  $c_1(T) = (r-s)H + c_1(Z) + c_1(V) + c_1(V')$  by [13, (32)].

The  $J$ -function of a smooth projective  $Y$  is  $J^Y(Q, t, z) = zS^Y(Q, t, z)^{-1}\mathbf{1}$  as in [13, (20)]; it satisfies

$$J^Y(Q, t, z) = z\mathbf{1} + t + O(z^{-1}). \quad (2)$$

The *extremal* specialization [13, (28) and (33)] sets  $Q^d = 1$  for  $d = k[L]$  with  $[L]$  the class of a line in a fibre of  $\mathbf{P}(V) \subset T \rightarrow Z$ , sets  $Q^d = 0$  for all other  $d$ , and replaces the parameter by

$$\bar{t} = t + \ln(q)c_1(T), \quad t \in H^*(Z). \quad (3)$$

In the specialized variables (2) reads  $J^T(q, t, z) = z\mathbf{1} + \bar{t} + O(z^{-1})$ , and the prefactor  $ze^{t/z}q^{c_1(T)/z} = ze^{\bar{t}/z}$  of (1) is exactly the  $d = 0$  summand. Whether (1) is the  $J$ -function is therefore the question of whether the summands with  $d \geq 1$  contribute in  $z$ -order  $\geq 0$ .

Three statements of [13] carry a hypothesis on  $\nu = r - s$ , and the whole of Sections 5–9 depends on them through the chain recorded below. We list each with the inequality it states and the statements that use it.

| Location in [13]       | Hypothesis     | Used by                                   |
|------------------------|----------------|-------------------------------------------|
| Theorem 4.4            | $r - s > 1$    | (37); Theorem 4.6; (39)–(40)              |
| Remark 4.5(3)          | $r - s \leq 1$ | nothing; a negative assertion             |
| Section 7, before (64) | $r - s > 1$    | (64); Proposition 7.5 and its corollaries |

The first row is the load-bearing one. [13, Theorem 4.6] deduces the presentation

$$QH_{\text{ext}}^*(T) = H^*(Z)[H][[q]] / \left\langle \prod_{i=1}^r (\rho_i + H) = q^{r-s} \prod_{j=1}^s (\sigma_j - H) \right\rangle, \quad (4)$$

which is [13, (37)], from (1) by exhibiting a quantum differential operator annihilating  $J^T$  and quoting [6, Theorem 10.3.1] and [12, Lemma 2]. That argument uses  $r - s > 1$  only through [13, Theorem 4.4]. The spectral computations [13, Propositions 5.1 and 5.2, Corollary 5.3] then use (4) and the ring homomorphism of [13, Theorem 3.3], and carry no hypothesis on  $\nu$  of their own; the inequality in [13, Theorem 1.2] is inherited from [13, Theorem 4.4]. Likewise, the modified extremal  $J$ -function [13, (40)] used throughout Sections 7–9 is derived from (1), and [13, Proposition 9.1] uses (4) again. Establishing (1) at  $\nu = 1$  therefore unblocks all of this at once.

The second row asserts the opposite of what we prove, and is used nowhere: [13, Remark 4.5(3)] is explicitly marked as not needed, and the same is true of its counterpart [13, Remark 4.7] for  $T'$ . What we do use from it is its positive half, the statement that (1) lies on Givental's overruled Lagrangian cone of  $T$  for  $r - s \leq 1$ , which the authors obtain from [4], [5] and [13, Lemma 9.6].

The third row governs the aperture of the Barnes expansion only. Nothing in Sections 8 and 9 of [13] constrains  $\nu$ : the sector [13, (78)] of [13, Proposition 8.2] comes from [13, Theorem A.2], whose hypothesis  $|\arg t| < (\frac{r}{2} + 1)\pi$  carries no  $\epsilon$  and is valid at  $\nu = 1$ ; and the reductions of [13, Sections 9.1–9.4] — weak-to-strong in the local model, the reduction to the split case, the cohomological reduction to the local model, and the comparison of the Fourier–Mukai transform with the local model — are stated and proved with  $r > s$  as the only inequality.

### 3 $J$ -normalization at discrepancy one

By the splitting principle of [13, Corollary 4.3], we may pull back injectively to a common flag bundle over  $Z$  on which both  $V$  and  $V'$  have filtrations by line bundles. The classes  $\rho_i$  and  $\sigma_j$  then exist individually; we retain the same notation after pullback. Write

$$A = H^*(\mathbf{P}(V)) \cong H^*(Z)[H] / \langle \prod_{i=1}^r (\rho_i + H) \rangle$$

for the coefficient ring of (1). It is a finite-dimensional graded  $\mathbf{C}$ -algebra whose elements of positive degree are nilpotent. Each factor of the denominator of (1) has the form  $mz + a$  with  $m \geq 1$  and  $a = \rho_i + H \in A$ , so

$$\frac{1}{mz + a} = \frac{1}{mz} \sum_{l \geq 0} \left( \frac{-a}{mz} \right)^l \in z^{-1} A[z^{-1}]$$

is a finite sum, and (1) is an element of  $zA[z^{-1}][[q]]$ . By the  $z$ -order of such an element we mean the largest  $n$  with a nonzero  $z^n$  coefficient. Since the pullback is injective coefficient-wise, the resulting order bound descends to the original cohomology ring; its symmetric expression is independent of the splitting.

**Lemma 3.1.** *For every  $d \geq 1$ , the degree- $d$  summand of (1) has  $z$ -order at most*

$$1 + s(d - 1) - rd = 1 - s - \nu d.$$

*Proof.* The numerator  $\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)$  has  $sd$  factors. The  $s$  factors with  $m = 0$  are  $\sigma_j - H$  and contain no  $z$ ; each of the remaining  $s(d - 1)$  factors is linear in  $z$ . So the numerator is a polynomial in  $z$  over  $A$  of degree at most  $s(d - 1)$ . The denominator  $\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)$  has  $rd$  factors, all with  $m \geq 1$ , so by the expansion above its inverse lies in  $z^{-rd} A[z^{-1}]$ . Multiplying by the leading factor  $z$  of (1) adds one.  $\square$

The two hypotheses of Theorem 1.1 enter here and nowhere else.

**Corollary 3.2.** *If  $\nu = 1$  and  $s \geq 1$ , then every summand of (1) with  $d \geq 1$  has  $z$ -order at most  $1 - s - d \leq -1$ , and hence*

$$z e^{t/z} q^{c_1(T)/z} \sum_{d \geq 0} q^{(r-s)d} \frac{\prod_{j=1}^s \prod_{m=0}^{d-1} (\sigma_j - H - mz)}{\prod_{i=1}^r \prod_{m=1}^d (\rho_i + H + mz)} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1}),$$

with  $\bar{\mathbf{t}} = t + \ln(q)c_1(T)$  as in (3).

*Proof.* The bound of Lemma 3.1 is  $1 - s - d$ , which is at most  $-1$  for  $d \geq 1$  and  $s \geq 1$ . The  $d = 0$  summand is  $z e^{t/z} q^{c_1(T)/z} = z e^{\bar{\mathbf{t}}/z} = z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ .  $\square$

**Remark 3.3.** Both hypotheses are sharp. For  $\nu = 1$  and  $s = 0$ , so  $(r, s) = (1, 0)$ , the bound of Lemma 3.1 is  $1 - d$  and is attained by the formal  $d = 1$  term of the hypergeometric expression:

$$z \cdot \frac{q}{\rho_1 + H + z} = q\mathbf{1} - \frac{q(\rho_1 + H)}{z} + \dots$$

Its  $z^0$  coefficient is  $q\mathbf{1}$ , so this formal expression is not  $J$ -normalized. Geometrically  $\mathbf{P}(V) = Z$  has point fibres and no line class  $[L]$ , so  $(1, 0)$  is a degenerate endpoint of the extremal-line setup, not a further standard-flip case repaired here. For  $\nu = 0$  the bound is the  $d$ -independent value  $1 - s$  and the sum is not  $q$ -adically convergent, which is why  $r > s$  is assumed throughout [13].

To convert the normalization into an identification we use the standard characterization of the  $J$ -function as a slice of Givental's cone. Let  $\mathcal{L}_T$  denote the overruled Lagrangian cone of the genus-zero Gromov–Witten theory of  $T$ , in the extremal specialization (3). The specialization is legitimate for the cone formalism because  $q$  carries positive weight: by [13, (33)] the curve class  $k[L]$  contributes  $q^{(r-s)k}$ , and  $r > s$ , so  $\mathcal{L}_T$  is defined over  $\mathbf{C}[[q]]$  and (1) is a  $q$ -adically convergent point of the ambient space.

**Lemma 3.4** ( $J$ -slice uniqueness; [8], [5]). *Let  $f \in \mathcal{L}_T$  be of the form  $f = z\mathbf{1} + \tau + O(z^{-1})$  for some  $\tau$  in the parameter space. Then  $f = J^T(\tau, z)$ .*

This is the mechanism behind every mirror theorem of Givental type: a point of the cone with a prescribed  $z^0$  coefficient is the  $J$ -function at that coefficient, and the mirror map is precisely the discrepancy between the prescribed  $z^0$  coefficient and the parameter one started with.

**Proposition 3.5.** *Let  $r = s + 1$  with  $s \geq 1$ . Then the right-hand side of (1) equals the extremal  $J$ -function  $J^T(q, t, z)$  of the local model, in the variables and completion of [13, Theorem 4.4].*

*Proof.* By [13, Remark 4.5(3)] the right-hand side of (1) lies on  $\mathcal{L}_T$  whenever  $r - s \leq 1$ ; the authors deduce this from [4], [5] and [13, Lemma 9.6]. By Corollary 3.2 it has the form  $z\mathbf{1} + \bar{\mathbf{t}} + O(z^{-1})$ . Lemma 3.4 with  $\tau = \bar{\mathbf{t}}$  gives the claim.  $\square$

**Corollary 3.6.** *Let  $r = s + 1$  with  $s \geq 1$ . Then the presentation (4) of [13, Theorem 4.6] holds, the modified extremal  $J$ -function [13, (39)–(40)] of  $T$  is as printed, and [13, Theorem 1.2] holds: after the extremal specialization,  $c_1(X) \star_q -$  on  $QH^*(X)$  has eigenvalue 0 with multiplicity  $\text{rank } H^*(X')$  and the single nonzero eigenvalue*

$$\lambda_0(q) = e^{-\pi\sqrt{-1}s}q = (-1)^sq$$

*with multiplicity  $\text{rank } H^*(Z)$ .*

*Proof.* The proof of [13, Theorem 4.6] specializes the parameter  $t$  of [13, Theorem 4.4], exhibits a differential operator annihilating  $J^T$ , and concludes with [6, Theorem 10.3.1] and [12, Lemma 2]; it uses  $r - s > 1$  only to know that (1) is  $J^T$ . Proposition 3.5 supplies that, so the proof applies verbatim, and [13, (39)–(40)] follow by the same substitution as in [13, Section 4.3].

For the spectrum, [13, Propositions 5.1 and 5.2] concern the abstract ring (4) and its subspaces and assume only  $r > s$ , and [13, Corollary 5.3] combines them with the ring homomorphism  $i^*: QH_{\text{ext}}^*(X) \rightarrow QH_{\text{ext}}^*(T)$  of [13, Theorem 3.3]. Once (4) is available at  $\nu = 1$ , that argument is unchanged. Its output is  $r - s$  nonzero eigenvalues  $\lambda_k(q) = (r - s)e^{-\pi\sqrt{-1}(2k+s)/(r-s)}q$  for  $0 \leq k < r - s$ ; at  $r - s = 1$  only  $k = 0$  occurs, and  $\lambda_0(q) = e^{-\pi\sqrt{-1}s}q$ .  $\square$

**Remark 3.7.** The companion statement [13, (34)] for  $\mathbf{P}(V)$  is the  $s = 0$  case of the same expression, and Lemma 3.1 applied to it gives maximal  $z$ -order  $1 - rd$ . This is at most  $-1$  for every  $d \geq 1$  as soon as  $r \geq 2$ , which holds whenever  $r = s + 1$  with  $s \geq 1$ . So [13, (34), (36) and (38)] are covered in the same range, by the same count.

Proposition 3.5 and Corollary 3.6 prove Theorem 1.1. They also settle the status of [13, Remark 4.5(3)]: under the standing assumption  $r > s$  its range  $r - s \leq 1$  is exactly  $\nu = 1$ , and by Corollary 3.2 and Remark 3.3 its negative assertion holds there only for  $s = 0$ .

## 4 The Meijer aperture at $\nu = 1$

The asymptotic analysis of [13, Section 7] rests on [13, Lemma 7.4], which writes the equivariant central-charge kernel  $\kappa_m$  as a cohomology-valued Meijer  $G$ -function,

$$\kappa_m = e^{-\pi\sqrt{-1}c_1(V')} G_{s,r}^{r,0} \left( \begin{matrix} 1 - \sigma_1, \dots, 1 - \sigma_s \\ \rho_1, \dots, \rho_r \end{matrix} \middle| t_m \right), \quad t_m := e^{-\pi\sqrt{-1}(2m+s)} \left( \frac{q}{z} \right)^{r-s}. \quad (5)$$

In the notation of [13, Appendix A] the lower and upper index counts of this  $G$ -function are  $s$  and  $r$ , so the parameter governing the asymptotic expansion is

$$\nu = r - s, \quad (6)$$

which is the discrepancy of the flip. (The appendix of [13] names the index counts  $p$  and  $q$ ; we write  $s$  and  $r$  to keep  $q$  for the extremal variable.)

[13, Theorem A.1] is Barnes' asymptotic expansion [2] in the form given by Meijer [10] and Luke [9]. Its hypothesis is stated there as

$$|\arg t| < (\nu + \epsilon)\pi, \quad \epsilon = 1 \text{ if } \nu > 1, \quad \epsilon = \frac{1}{2} \text{ if } \nu = 1. \quad (7)$$

The  $\epsilon$  of (7) is the whole issue. The expansion printed in [13, Section 7] after [13, Lemma 7.4] is introduced by the sentence “For  $r - s > 1$ ”, and the sector recorded with it,

$$\left| \arg \left( \frac{z}{q} \right) + \frac{(2m+s)\pi}{r-s} \right| < \left( 1 + \frac{1}{r-s} \right) \pi, \quad (8)$$

which is [13, (64)], is exactly (7) with  $\epsilon = 1$  transported through (5).

**Lemma 4.1.** *Let  $\nu = 1$ . Applied to (5), [13, Theorem A.1] gives the asymptotic expansion of [13, Proposition 7.5] on*

$$\left| \arg \left( \frac{z}{q} \right) + (2m+s)\pi \right| < \frac{3\pi}{2}. \quad (9)$$

*The sector obtained by substituting  $r - s = 1$  into (8) is  $|\arg(z/q) + (2m+s)\pi| < 2\pi$ , wider by  $\pi/2$  on each side.*

*Proof.* From (5),  $\arg t_m = -(2m+s)\pi - \nu \arg(z/q)$ . At  $\nu = 1$  the hypothesis (7) reads  $|\arg t_m| < (1 + \frac{1}{2})\pi$ , which is (9). Substituting  $r - s = 1$  into (8) gives  $(1+1)\pi = 2\pi$ , the value of  $(\nu + \epsilon)\pi$  at  $\epsilon = 1$ .  $\square$

The correction is one-directional: (9) is smaller than the naive specialization of (8), so a repair must show that what remains is still enough. The other sector in play is unaffected.

**Lemma 4.2.** *[13, Proposition 8.2] and its sector*

$$\left| \arg \left( \frac{z}{q} \right) - \frac{(1-s)\pi}{r-s} \right| < \frac{\pi}{2} + \frac{\pi}{r-s}, \quad (10)$$

*which is [13, (78)], are valid at  $\nu = 1$ , where (10) reads  $|\arg(z/q) - (1-s)\pi| < \frac{3\pi}{2}$ .*

*Proof.* [13, Proposition 8.2] is deduced from [13, (76)], which applies [13, Theorem A.2] to a  $G_{s,r}^{r,1}$  function of the same argument. The hypothesis of [13, Theorem A.2] is  $|\arg t| < (\frac{\nu}{2} + 1)\pi$ ; it involves no  $\epsilon$  and no restriction on  $\nu$ . Transporting it through the same substitution as above gives (10) for every  $\nu \geq 1$ , and the displayed form at  $\nu = 1$ .  $\square$

**Proposition 4.3.** *Let  $r = s + 1$  with  $s \geq 1$  and let  $0 \leq k \leq 1$ , so that the semiorthogonal decomposition [13, Theorem 1.1] has the single  $Z$ -component of index  $m = -k$ . The sectors (9) and (10) meet in the open sector*

$$\max((2k - s)\pi, (1 - s)\pi) - \frac{3\pi}{2} < \arg\left(\frac{z}{q}\right) < \min((2k - s)\pi, (1 - s)\pi) + \frac{3\pi}{2}, \quad (11)$$

*of opening  $2\pi$ . It contains the eigenvalue ray  $\arg(z/q) = (2k - s)\pi$  of [13, Theorem 1.4(1)] and the tame ray  $\arg(z/q) = (1 - s)\pi$  of [13, Theorem 1.4(2)].*

*Proof.* With  $m = -k$ , sector (9) is the open interval of half-width  $3\pi/2$  about  $(2k - s)\pi$ , and sector (10) is the open interval of half-width  $3\pi/2$  about  $(1 - s)\pi$ . For  $k \in \{0, 1\}$  the two centres are at distance  $|2k - 1|\pi = \pi$ , so the intervals meet in an interval of length  $3\pi - \pi = 2\pi$ , namely (11). Since  $\pi < 3\pi/2$ , each centre lies in the other interval, so both centres lie in the intersection; they are the two rays named in the statement, because at  $r - s = 1$  the rays  $-(2m + s)\pi/(r - s)$  and  $(1 - s)\pi/(r - s)$  of [13, Theorem 1.4] are  $(2k - s)\pi$  and  $(1 - s)\pi$ .  $\square$

Lemmas 4.1 and 4.2 and Proposition 4.3 prove Theorem 1.2.

*Remark 4.4.* For a blow-up,  $s = 1$ , the numbers are: (9) with  $k = m = 0$  is  $-\frac{5\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ ; (10) is  $|\arg(z/q)| < \frac{3\pi}{2}$ ; and (11) is  $-\frac{3\pi}{2} < \arg(z/q) < \frac{\pi}{2}$ , containing the tame ray 0 and the eigenvalue ray  $-\pi$ . The codimension-two blow-up is  $(r, s) = (2, 1)$ , where  $\lambda_0(q) = -q$  and the exponential factor of the expansion of [13, Proposition 7.5] is  $\exp(-\lambda_0(q)/z) = \exp(q/z)$ .

*Remark 4.5.* The opening  $2\pi$  of (11) exceeds  $\pi$ , the width beyond which an asymptotic expansion of level one determines its flat section uniquely; see [1]. The oriented comparison of the nonzero-eigenvalue and tame expansions carried out in [13, Section 9], which needs both expansions to be valid simultaneously on a sector containing both rays, is therefore available at  $\nu = 1$ . In particular the common-sector condition of [13, Remark 1.6], namely  $\frac{1}{4}(r - s - 6) < k < \frac{3}{4}(r - s + 2)$ , which at  $r - s = 1$  reads  $-\frac{5}{4} < k < \frac{9}{4}$  and admits both  $k = 0$  and  $k = 1$ , holds at  $\nu = 1$  for the corrected sector (9) and not only for (8).

## 5 What is repaired, and what is not

**Theorem 5.1.** *Let  $X \dashrightarrow X'$  be a standard flip with  $r = s + 1$  and  $s \geq 1$ . Then [13, Theorems 1.2, 1.4, 9.9 and 9.14] and [13, Corollary 1.5] hold for  $X$ . In particular they hold for the blow-up of a smooth projective variety along a smooth centre of codimension two, the case  $(r, s) = (2, 1)$ .*

*Proof.* Corollary 3.6 gives (4) and [13, Theorem 1.2], and makes [13, (39)–(40)] available, so the modified extremal  $J$ -function used from [13, Section 7] onward is the one the argument assumes. [13, Proposition 7.5], [13, Corollary 7.6] and [13, Proposition 7.8] then hold with (8) replaced by (9), by Lemma 4.1; [13, Propositions 8.2 and 8.3] and [13, Corollary 8.4] hold as printed, by Lemma 4.2.

The reductions of [13, Sections 9.1–9.4] are then unchanged. [13, Proposition 9.1] uses the presentation (4) and the extremal Dubrovin connection; [13, Proposition 9.2] combines it with [13, Propositions 7.8 and 8.4]; [13, Corollary 9.3] takes a partial nonequivariant limit; [13, Section 9.2] removes the splitting assumption on  $V, V'$ , giving [13, Corollary 9.7]; [13, Section 9.3] passes from the local model to  $X$ , giving [13, Theorem 9.9]; and [13, Section 9.4] compares the Fourier–Mukai transform with the local model, giving [13, Theorem 9.14]. None of these steps constrains  $\nu$  beyond  $r > s$ . The two halves of [13, Theorem 1.4] are

[13, Theorems 9.9 and 9.14], and [13, Corollary 1.5] is the translation of [13, Theorem 1.4] through the functors [13, (3)]. Proposition 4.3 supplies the sector on which both expansions used in that translation are valid.  $\square$

Three boundaries should be stated explicitly.

First,  $(r, s) = (1, 0)$  is not covered. By Remark 3.3 the formal expression (1) has a  $z$ -order-zero term there and is not  $J$ -normalized. This is the rank-one projective-bundle endpoint, where  $\mathbf{P}(V) = Z$ ,  $X' = \emptyset$ , and the point fibres contain no line class  $[L]$ . It therefore lies outside both the normalization argument and the geometric extremal-line setup. All genuine discrepancy-one standard flips have  $s \geq 1$  and are covered by Theorem 5.1.

Second, everything above lives on the extremal slice. The Novikov parameter is restricted to multiples of the class  $[L]$  of a line in a fibre of  $\mathbf{P}(V)$ , all other Novikov variables are zero, and the parameter is  $\bar{t} = t + \ln(q)c_1(T)$  with  $t$  pulled back from  $H^*(Z)$ ; this is the specialization in which [13, Theorems 1.2 and 1.4] are stated, and no statement here concerns the unrestricted quantum product or a comparison across the full Novikov ring.

Third, the two corrections are local to the two hypotheses named in Section 2. For  $\nu > 1$  nothing changes: the argument of [13] applies as printed, with (8) supplied by the  $\epsilon = 1$  branch of [13, Theorem A.1]. Every other ingredient of [13] — the ring homomorphism to the local model, the spectral computation in the abstract presentation, the deformation to the normal cone, the Fourier–Mukai comparison — is used exactly as it stands.

The count in Lemma 3.1 is what makes the discrepancy-one range behave like the higher-discrepancy one. The  $s$  numerator factors  $\sigma_j - H$  at  $m = 0$ , free of  $z$ , are worth exactly  $s$  powers, and at  $\nu = 1$  they are the entire margin: the maximal order  $1 - s - \nu d$  sits at  $-1$  precisely when  $s = 1$  and  $d = 1$ , and rises to 0 the moment  $s$  drops to zero. So the boundary between the identification  $I = J$  and a nontrivial mirror map at  $\nu = 1$  is not the codimension of the centre but the presence of at least one factor in  $V'$ .

## AI assistance disclosure

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